

Answer Key

Grade 6 Mathematics • Unit 4: Working with Percents • Answers for every item on Sheets 1 and 2 of each lesson packet. Constructed responses show a model answer; accept equivalent reasoning.

LESSON 4-1

Percent Means Per Hundred • Percents Greater Than 100%

Sheet 1 • Percent Means Per Hundred

1. a. 46% b. 54%
2. a. 30% b. 28% c. 50% d. 34% e. 40% f. 15% g. 52% h. 90%
3. Missing: 27, 36. 9 out of 25 = 36 out of 100 = 36%.
4. a. $22 \times 2 = 44$ out of 100 = 44% b. $100\% - 44\% = 56\%$
5. Equivalent ratios are made by multiplying (or dividing) both terms by the same number, not adding. $20 \times 5 = 100$, so $7 \times 5 = 35$: 7 out of 20 = 35%.

Sheet 2 • Percents Greater Than 100%

1. a. 300% b. 175%
2. Missing percents: 50%, 75%, 125%, 150%, 175%. Tuesday = 175% of Monday.
3. 260 to 200 = 130 to 100, so 130% (more than the goal).
4. 33 to 20 = 165 to 100 (multiply both by 5), so 165%.
5. 60% of 30 cm is only 18 cm, which is less than 30 cm. 60 is 2 times 30, so 60 cm is 200% of 30 cm.

TEST PRACTICE

6. D – 150%
7. B – 2.5 times

LESSON 4-2

Fractions, Decimals, Percents • Compare and Order

Sheet 1 • Fractions, Decimals, Percents

1. a. $35/100 = 7/20$ b. 0.35 c. 35%
2. $1/4 = 0.25 = 25\%$; $7/10 = 0.7 = 70\%$; $2/25 = 0.08 = 8\%$; $13/20 = 0.65 = 65\%$; $3/5 = 0.6 = 60\%$
3. $11/25 = 44/100 = 0.44 = 44\%$
4. She just wrote the digits 3 and 4 side by side; that is not the same value. $3/4 = 75/100 = 0.75 = 75\%$.

Sheet 2 • Compare and Order

1. a. $>$ b. $<$ c. $<$
2. a. $7/10$, 0.72, 75% b. 0.2, 22%, $1/4$ c. $6/5$, 125%, 1.3
3. A = 55%, B = 60%, C = 52%. Order: Room C, Room A, Room B (Room B greatest).
4. $3/5 = 60\%$ and $0.7 = 70\%$, so any value between 60% and 70%. Sample: $13/20 = 0.65 = 65\%$.
5. Comparing single digits does not compare values; change to one form. $3/8 = 0.375 = 37.5\%$ and $0.4 = 40\%$, so $3/8 < 0.4$.

TEST PRACTICE

6. C – 0.35, 38%, $2/5$ (35%, 38%, 40%)
7. C – 64%

LESSON 4-3

Estimate with Benchmarks • Estimate a Percent of a Number

Sheet 1 • Estimate with Benchmarks

1. Pages: 40, 80, 120. 42 is close to 40, so about 25%.
2. a. 10% b. 75% c. 25% d. 50% e. 100% f. 25% g. 10% h. 75%
3. a. \$50 b. 20% of 500 = \$100, and \$96 is close to \$100, so about 20%.
4. 75% of 48 = 36, and 35 is close to 36, so about 75%.
5. 26 out of 50: half of 50 is 25, so it is just over 50% (exactly 52%). 26 out of 60: half of 60 is 30, so it is below 50% (about 43%). 26 out of 50 is closer to 50%.

Sheet 2 • Estimate a Percent of a Number

1. a. about 6 (10% of 60) b. about 30 (75% of 40) c. about 50 (25% of 200)
2. a. \$20 b. about \$60 ($3 \times \$20$)
3. 50% of 400 = 200, so about 200 students.
4. 25% of 800 = $800 \div 4 = 200$, so about 200 apples.
5. 20% is not 20; she multiplied by 20 instead of 0.20. 10% of 500 = 50, so 20% of 500 = 100. 19% of 512 is about 100.

TEST PRACTICE

6. B – 20 (25% of 80)
7. C – \$48 (40% of \$120)

LESSON 4-4*Find the Percent of a Number · Compare Percents***Sheet 1 · Find the Percent of a Number**

1. a. 9 b. 48 c. 21 d. 36 e. 45 f. 30
2. Missing: 72, 108, 144. 40% of 180 = 72 students.
3. a. $0.25 \times 64 = \$16$ b. $\$64 - \$16 = \$48$
4. 10% of 40 = 4 and $5\% = 2$, so $15\% = 6$ players ($0.15 \times 40 = 6$).
5. They are equal: $0.3 \times 90 = 27$ and $0.9 \times 30 = 27$. Both equal $30 \times 90 \div 100$, so switching the numbers gives the same amount.

Sheet 2 · Compare Percents

1. a. 70% b. 65% c. 75%
2. Ava 45%, Ben 48%, Cruz 44%. Ben has the best record.
3. Rosa: $34/40 = 85/100 = 85\%$. Kai: $42/50 = 84/100 = 84\%$. Rosa did better, even though Kai spelled more words.
4. A, C (75% and 72%; 17 out of 25 = 68%, 7 out of 10 = 70% exactly)
5. Equal rates come from multiplying, not adding the same amount. $18/24 = 3/4 = 75\%$ and $24/30 = 4/5 = 80\%$, so the rates are not equal; 24 out of 30 is better.

TEST PRACTICE

6. C – 33 out of 50 = 66% (65%, 64%, 60%)
7. B – Team A: $12/16 = 75\%$; Team B: $15/25 = 60\%$; Team A did better.

LESSON 4-5*Find the Whole · Solve for the Whole***Sheet 1 · Find the Whole**

1. $20\% = \$18$; dollars: 18, 36, 54, 72, 90. The helmet costs \$90.
2. a. 130 b. 44 c. 70 d. 80 e. 140 f. 90
3. One part (25%) = $18 \div 3 = 6$ gal; full = $4 \times 6 = 24$ gallons.
4. $25\% = 120 \div 3 = 40$ pages, so $100\% = 4 \times 40 = 160$ pages.
5. A is greater: 20 is a smaller percent (40%) of A, so A must be bigger. $A = 50$ ($10\% = 5$) and $B = 25$ ($10\% = 2.5$, or $20 \div 0.8$).

Sheet 2 · Solve for the Whole

1. a. 70 b. 48 c. 80
2. a. $0.55 \times w = 33$ b. $w = 33 \div 0.55 = 60$
3. $0.75 \times w = 315$, $w = 315 \div 0.75 = \$420$. Still needed: $\$420 - \$315 = \$105$.
4. $0.35 \times w = 42$, $w = 42 \div 0.35 = 4,200 \div 35 = \120 . Check: $0.35 \times 120 = 42$.
5. He divided by 35 instead of 0.35 ($35\% = 0.35$); a whole of 0.8 is smaller than the part, which makes no sense. $w = 28 \div 0.35 = 80$. Check: $0.35 \times 80 = 28$.

TEST PRACTICE

6. B – $0.3 \times w = 45$
7. C – 30